

Sales Performance Analysis: An End-to-End Data Project

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Project Introduction

As part of my BCA studies, I wanted to build a project that mirrors a real-world business scenario. I took a raw sales dataset and moved it through the entire data lifecycle- from messy CSV files to a clean, interactive dashboard. The goal was to answer a simple question: *“How can we turn thousands of rows of shipping data into a clear strategy for a business manager?”*

The Tech Stack

- **Python (Pandas):** My primary tool for data “wrangling”.
- **Power BI:** Used for building the visual interface and complex calculations.
- **Github:** For documenting my progress and hosting the final results.

My Process (Behind-the-Scenes)

Phase 1: Cleaning the “Mess”

Raw Data is rarely perfect. I used Python to audit the dataset. I found several columns with missing values (like STATE and TERRITORY) and addressed them. I also realized the dates were stored as text, which made it impossible to see trends. I converted them into a proper timeline format so I could track sales growth over months and years.

Phase 2: Defining the Metrics

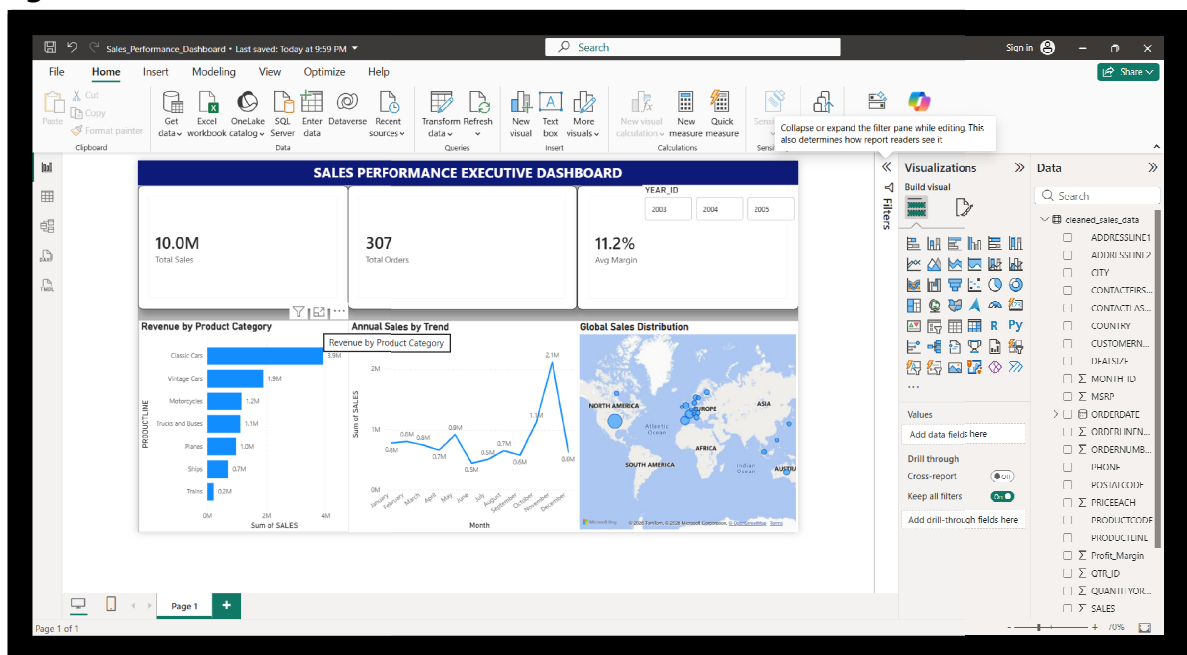
In Power BI, I didn't just want to show "Sum of Sales." I created specific measures to give better context:

- **Total Revenue:** The big-picture number (\$10.0M).
- **Profit Margin:** I calculated this to see how much of our sales were actually translating to profit (11.2%).

Phase 3: Building for the User

I designed the dashboard to be "Executive Ready." I placed the most important numbers (KPIs) at the very top so they are the first thing a manager sees. I also added **Yearly Buttons (Slicers)** so that instead of looking at a static report, a user can "interact" with the data to see how 2004 is compared to 2005.

Figure 1: Final Interactive Sales Dashboard



What I Discovered

- **The “Star” Product:** “Classic Cars” is by far our most successful category.
- **Timing is everything:** There was a significant peak in sales during 2004, which gives us a baseline for future goals.
- **Regional Dominance:** While our market is global, the USA and Europe remain our strongest revenue pillars.

Final Thoughts

This project taught me that data analysis isn't just about the charts- it's about the cleaning and logic that happens before the first chart is even built. It has greatly improved my skills in both Python and Power BI.